

Project: Smart Parking Platform

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I. Research: Comparative Analysis

The Smart Parking Platform will allow drivers to search for and find parking and then reserve and pay for parking in real time. Some applications have already implemented a handful of these features, but none combined all the most important aspects of parking into a single integrative payment and navigation system. ParkMobile is a mobile payment system, as well to extend your parking time without running back to the meter.

Still, it is largely an entirely payment-oriented system with limited real-time garage info & reservations. Our system will offer all four of these features.

Spot Hero offers the ability to search for and book parking in advance, mainly near events, airports, and high-traffic urban locations. Although the model is first and foremost operator-focused, it lacks comprehensive real-time occupancy tools. Our approach will benefit both drivers and operators. Park Whiz allows users to compare and place reservations at parking lots by location and price.

Our challenge is a lack of comprehensive, real time space tracking, live directions assistance, and reporting analytics. Our service offers real-time availability directions, receipts, and reporting. Passport Parking is very good for the mobile payment side of parking (used by cities, universities, and parking operators) but has not been very focused on finding free garage spaces or reserving spaces. We will include search reservation maps, alerts and management systems too.

Google Maps parking functionalities are designed to provide parking information to the car driver with search of parking places and navigation. Google Maps is not designed for the reservations and payment of parking. Our system will reuse the navigation support but will add reservation payment receipts, availability dashboard and operator dashboard functionalities.

II. Vision and Scope

The purpose of the Smart Parking Platform (SPP) is to develop a web-based and mobile-based application system that enables the drivers to locate, reserve, navigate and make payments for parking efficiently. The need for the business is that the drivers spend valuable time to search parking space, and because of this the situation causes aggravation and increases congestion.

The other business need is for parking operators to have more appropriate tools to monitor the parking occupancy, control the parking entrance fees, and increase the utilization of the parking facilities. Project scope encompasses these features and functionalities: user registration and login, live parking space online, map overlay with interactive parking location reservation online payment, reservation history, parking receipts notifications external maps overlay, administration

Significant project stakeholders encompass drivers, parking and transportation facility management, city transportation authorities, system administrators and operators, billing system administrators, their billing staff, third-party gateway via partner payment agency, development teams of all applications involved including the billing system. Considerations are: semester project's fixed scope, schedule and budget.

III. Drafts for Software Requirements Specification:

3.1 – Introduction

This document defines the initial software requirements document for the Smart Parking Platform. The system will cater for both the needs of parking drivers and parking operators in occupancy management, reservations management, fee payments and reporting needs

3.2 - Overall Description

The Smart Parking Platform will be offered in two formats: Web-based application and mobile based application. Drivers will use the system to identify reserve go-to and pay and report parking. Parking operators will use the system for overseeing parking limited capacity and its utilization, generate reports, provide data for estimating parking capacity and other predictive data and manage parking facility details. Users: Drivers, parking operators, city transportation departments, system administrators, billing and payment system administrators, service providers and developers. System Environment: Web based application and mobile based application. Constraints: semester system time constraints, third-party mapping and payment related API constraints, constraints related to security and privacy. Assumptions: motorists have internet access, the parking data is updated regularly, the navigation services and payments are available continuously and reliability.

3.3 – Functional Requirements

FR1: The system shall authorize the parking system based on login credentials provided by users

FR2: The system shall facilitate search of parking resources near a location of interest to a driver

FR3: The system shall provide real time parking display

FR4: The system shall allow reservation of parking resources by a driver

FR5: The system shall facilitate payments via digital means

FR6: The system shall generate a payment receipt and general reservation confirmation invoice for users.

FR7: The system shall establish notification and warning messages for users

FR8: The system shall enable search facility of parking occupancy reports by parking operators

3. 4 Non-Functional Requirements

3.4 – Non-Functional Requirements

NFR1: The system should be user-friendly both in web oriented and mobile oriented environments

NFR2: The system should provide expected degree of security and data safety among account and payment holders

NFR3: The parking resources data should be timely, accurately and consistently updated

NFR4: The system should be operational and reliable in peak cluster hours

NFR5: The system should rely on third-party API of navigation and payment processing

3.5 – Use Cases

1 Sign in and Sign Up system driver create a new account

2 log in system driver login process

3 search for parking system driver identifies parking interest

4 Get viewing of real time parking display system driver views parking now

5 Interactive viewing of parking system driver can view parking location on a map

6 search parking position with sequential filter system driver analyzes parking data with logical filter

7 Reserve parking space system driver works on parking request

8 Receive reservation confirmation system driver wait for parking notification, get notified

9 Direction to parking system driver uses navigation system

10. Pay system driver initiate to pay
11. Access reservation history system driver view of previously paid parking spaces
12. View payment receipt system driver view transportation number and date
13. notification parking system driver get reminded
14. log into admin dashboard system parking operators view1 parking details
15. View parking occupancy reports system parking operators view2 parking status

IV. Conclusion

The first version of the project document gives an overview of the Smart Parking Platform, makes a difference between the Smart Parking Platform, the parking solutions already present, starts to elaborate The Vision and Scope section and offers a draft of the Requirement Specification Document which features 15 use cases. Later revisions will include more requirements, a more detailed view of stakeholders, and thorough project planning.